

Exercise 1. Find the forced response, that is find $i(t)$, and draw the pole-zero plot of each transfer function $\frac{I}{V}$. See section 2.3.

- (a) when $v(t) = 10e^{-2t}\cos(2t + 5)$ and $\frac{d^2i}{dt^2} + 3\frac{di}{dt} + 2i = 2\frac{dv}{dt} + 4v$ (ans: $i(t) = 8.94e^{-2t}\cos(2t - 111.6)$)

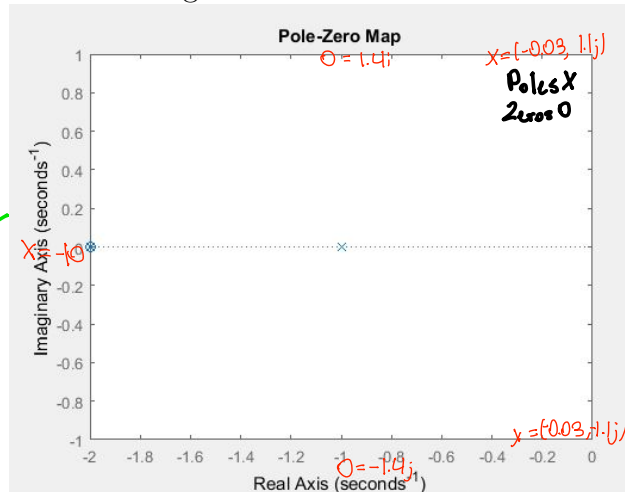
$$\frac{I}{V} = \frac{2s+4}{s^2+3s+2}$$

$$\frac{I}{V} = \vec{V} \cdot \frac{2s+4}{s^2+3s+2} \quad s = -2 + 2j$$

$$\vec{V} = 10 \angle 0 \quad \vec{I} = 8.94 \angle -111.6$$

$$i(t) = 8.94 e^{-2t} \cos(2t - 111.6) \quad \checkmark$$

Figure 1: Pole Zero Plot



b) $\frac{I}{V} = \vec{V} \cdot \frac{2s^2+4}{s^2+10s+2s+10}$

$\vec{V} = 10 \angle 0 \quad s = 4j$

$\vec{I} = 1.75 \angle -20.47$

$i(t) = 1.75 \cos(4t - 20.47)$

Poles: $\text{cZero}(s^2+10s+2s+10) = (-9.9, 0) \quad (-0.04, 1.1) \quad (-0.04, -1.1)$

Zeros: $\text{cZero}(2s^2+4) = (0, -1.4j) \quad (0, 1.4j)$

c) $v(t) = 10e^{-3t} \quad \frac{I}{V} = 2 \left(\frac{s-1}{s+2} \right) - \left(\frac{s-3}{s+5} \right)$

Poles: $-5, -2 \quad \vec{V} = 10 \angle 0 \quad s = -3$

Zeros: $-9.4, 0.42 \quad \vec{I} = 14.38 \angle 16.96$

$i(t) = 14.38 e^{-3t} \cos(16.96)$

- (b) when $v(t) = 10\cos(4t)$ and $\frac{d^3i}{dt^3} + 10\frac{d^2i}{dt^2} + 2\frac{di}{dt} + 10i = 2\frac{d^2v}{dt^2} + 4v$

- (c) when $v(t) = 10e^{-3t}$ and $\frac{I}{V} = 2 \left(\frac{s-1}{s+2} \right) - \left(\frac{s-3}{s+5} \right)$ hint: first make a single fraction

Exercise 2. Find the natural response of the transfer functions (either found or given) in question 1. That is find $i(t)$ when $v(t) = 0$ which is based on roots of the transfer function. Section 2.5.

b) $k_1 e^{-9.9t} + k_2 e^{-0.04t} \cos(1.1t + \theta)$

- (a) (ans: $K_1 e^{-2t} + K_1 e^{-1t}$) c) $k_1 e^{-5t} + k_2 e^{-2t}$

Exercise 3. Use MATLAB's roots() function to verify the poles and zeros of the transfer functions in question 1. The root() function will be done once for each numerator and each denominator. Print a screen shot of the command window.

- (a) ans:
zeros = roots([2 4])
poles = roots([1 3 2])

Exercise 4. Use MATLAB's polyval function (see lab 1) to evaluate $H = s^2 + 3s + 10$ when $s = 3$. Submit the MATLAB code and a screen shot of the command window.

Exercise 5. What will MATLAB produce for the following operations if $x=[3 \ 6 \ 9]$ and $y=[5 \ 3 \ 0]$? If the command generates an error describe the fix and the intended results.

- (a) $x.+y \rightarrow$ fix $x+y$, $+$ already does element wise math
 $\begin{matrix} 8 & 9 & 9 \end{matrix}$
- (b) $x*y \rightarrow$ fix $x.*y$, need to do element wise $*$ because its a vector
 $\begin{matrix} 15 & 18 & 0 \end{matrix}$
- (c) $x+y \rightarrow$ $\begin{matrix} 8 & 9 & 9 \end{matrix}$
- (d) $x./y \rightarrow$ $\begin{matrix} 0.6 & 2.0 & \text{Inf}(\%) \end{matrix}$
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